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Discrete Linear Canonical Transform on Graphs: Fast Sampling Set Selection Method

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ABSTRACT

With the flourishing development of graph signal processing, an increasing number of classical signal processing methods are being incorporated into this field, and the graph linear canonical transform (GLCT) is one such example. In this paper, we address the problem of signal sampling set selection in the GLCT domain based on the proposed GLCT sampling theory. We present a novel fast sampling method. Furthermore, we discuss the relationship between the proposed method and existing sampling set selection methods based on the GLCT spectrum. It is demonstrated that the proposed method considers GLCT spectrum information without the need for the eigendecomposition of the variation operator. Finally, the performance of the proposed method was validated through the selection of vertices, comparing results in terms of reconstruction error and recovery time, which demonstrated its superior efficacy.

CCS CONCEPTS

• **Mathematics of computing** → *Spectra of graphs*.

KEYWORDS

Graph signal processing, graph linear canonical transform, sampling set selection, fast sampling algorithm

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1 INTRODUCTION

With the continuous increase of various networks and big data, graph signal processing (GSP) has rapidly developed [1–4]. GSP is a theory and underlying graph topology for analyzing and processing vertex data in graphs. This theory allows us to effectively apply signal processing techniques to many practical problems,

such as social, transportation, brain, and sensor networks [5–8], by analogizing techniques used for audio, images, or time-domain signals in traditional signal processing. This paper considers the key topic of graph signal processing: sampling theory.

Graph signal sampling is a fundamental task for handling large data or complex structured data in the real world [1, 2]. Processing such raw data consumes a significant amount of system resources, including storage and computational resources, thus there is a strong need for a sampling version that captures most relevant information from the data. The graph Fourier transform (GFT) is a crucial tool in graph signal sampling that expands signals into eigenvectors of adjacency matrices defined by corresponding eigenvalues.

Although the GFT can perform various operations on real-world graph signals, it cannot capture the transformation process from vertex domain to spectral domain and cannot handle graph signals with chirp characteristics [9, 10], leading to problems of insufficient degrees of freedom, lack of flexibility, and underutilization of parameters. In our previous research [11], we extended linear canonical transform [12] to graphs and introduced graph linear canonical transform (GLCT). The GLCT exhibits strong adaptability and flexibility in signal processing with its three free parameters as an extension of Fourier transform and fractional Fourier transform. It can handle graph signals with chirp characteristics.

We discussed sampling theory in GLCT domain in [13] and presented some performance analysis on this method. Unfortunately, most of these methods have high computational complexity as they require eigenvalue decomposition to compute graph linear canonical bases or some of their vectors. In previous studies [14–16], several signal recovery algorithms and sampling strategies were used to selectively choose vertices one by one using deterministic methods to minimize or maximize the objective cost function. These method strategies all perform reconstruction with ideal filters under different optimality criteria.

To downgrade computational complexity and shorten time, we introduce the method of [17] into our GLCT sampling. This method first filters vertices to obtain a set of admissible vertices that are sufficiently far away from already selected vertices. Then, it selects an optimal vertex from the admissible set. Based on this method, we propose a fast graph signal sampling method using graph localization filter operators. In order to select sufficiently distant vertices, we use GLCT spectral filters instead of ideal filters. We summarize the close relationship between the proposed method and previous sampling methods based on graph linear typical frequencies [13]. In experiments, we compare execution time and prediction errors in

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Table 1: The symbols used in this paper: $\mathbf{x} \in \mathbb{R}^N$ is any vector, $\mathbf{X} \in \mathbb{R}^{N \times M}$, and $\mathbf{Y} \in \mathbb{R}^{N \times N}$ are arbitrary matrices, while $\mathcal{A}$ and $\mathcal{B}$ are sets

Symbol	Definition
$\mathbf{Y}^M$	The GLCT operator or bandlimited operator in the GLCT domain
$\mathbf{X}^M$	N rows and M columns of the GLCT operator or bandlimited operator
$\lambda_i(\mathbf{Y})$	The i th eigenvalues of matrix $\mathbf{Y}$ $\lambda_{\min}(\mathbf{Y}) = \lambda_0(\mathbf{Y}) \leq \dots \leq \lambda_{N-1}(\mathbf{Y}) = \lambda_{\max}(\mathbf{Y})$
$u_i(\mathbf{Y})$	Eigenvector corresponding to $\lambda_i(\mathbf{Y})$
$\sigma_i(\mathbf{X})$	The i th singular value of matrix $\mathbf{X}$ $\sigma_{\min}(\mathbf{X}) = \sigma_0(\mathbf{X}) \leq \dots \leq \sigma_{N-1}(\mathbf{X}) = \sigma_{\max}(\mathbf{X})$
$\det[\mathbf{Y}]$	Determinant of $\mathbf{Y}$ with $\det[\mathbf{Y}] = \prod_{i=0}^{N-1} \lambda_i(\mathbf{Y})$
$\text{tr}[\mathbf{Y}]$	Trace of $\mathbf{Y}$ with $\text{tr}[\mathbf{Y}] = \sum_{i=0}^{N-1} \lambda_i(\mathbf{Y})$
$ \mathbf{x} $	$\text{diag}(\text{sgn}(\mathbf{x})) \mathbf{x}$
$ \mathbf{X} $	$\text{sgn}(\mathbf{Y}) \odot \mathbf{Y}$ where $\odot$ means element-wise product
$ \mathcal{A} $	Number of elements in $\mathcal{A}$
$\ \mathbf{x}\ _2$	ℓ_2 norm of $\mathbf{x}$
$\ \mathbf{X}\ _2$	Spectral norm of $\mathbf{X}$ with $\ \mathbf{X}\ _2 = \sigma_{\max}(\mathbf{X})$
$\ \mathbf{X}\ _F$	Frobenius norm of $\mathbf{X}$ with $\ \mathbf{X}\ _F = \left(\text{tr}(\mathbf{X}^H \mathbf{X})\right)^{1/2}$ where $\mathbf{X}^H$ is conjugate transpose of $\mathbf{X}$
$\mathbf{x}_{\mathcal{A}}$	Extracting elements indexed by set $\mathcal{A}$ to obtain a subvector of $\mathbf{x}$
$\mathbf{X}_{\mathcal{A}\mathcal{B}}, (\mathbf{X})_{\mathcal{A}\mathcal{B}}$	Extracting rows indexed by set $\mathcal{A}$ and columns indexed by set $\mathcal{B}$ to obtain a submatrix of $\mathbf{X}$
$\mathbf{X}_{\mathcal{A}}, (\mathbf{X})_{\mathcal{A}}$	$\mathbf{X}_{\mathcal{A}\mathcal{A}}$
$\mathbf{x}^T, \mathbf{X}^T$	Transpose of $\mathbf{x}$ or $\mathbf{X}$
$\mathbf{x}^*, \mathbf{X}^*$	Conjugate of $\mathbf{x}$ or $\mathbf{X}$
$\mathbf{Y}^{-1}$	Inverse of $\mathbf{Y}$
$\mathbf{X}^\dagger$	Pseudo-inverse $\mathbf{X}$

GLCT domain to evaluate the performance of the proposed method. The proposed method is faster and exhibits better performance in GLCT domain compared to previous methods.

The remaining parts of this paper are organized as follows. Section 2 provides a brief introduction to the basic concepts of graph signals, the fundamental theory of GLCT, and its sampling theorem. Section 3 presents a new reconstruction method for graph signals in the GLCT domain and proposes a sampling set selection method based on graph localization filter operators. Section 4 elucidates the profound relationship between the proposed method and previous methods, representing them in the form of localization filter operators as well. Section 5 demonstrates the performance of the proposed method compared to other methods in terms of execution time and recovery error through experimental results. Finally, Section 6 summarizes the entire paper and Table 1 summarizes most of the notations used in this paper.

2 PRELIMINARIES

In this section, we briefly review the concept of discrete signal processing in the GLCT domain [11] and its basic sampling theory [13].

2.1 Basic Concepts of Graph Signals

The graph $\mathcal{G}$ is denoted as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ and $\mathcal{E}$ denote the set of vertices and edges respectively. The number of vertices is given by $N = |\mathcal{V}|$, and $\mathcal{V} = \{v_0, \dots, v_{N-1}\}$. For the weighted adjacency matrix $\mathbf{A} \in \mathbb{C}^{N \times N}$ of the graph, the edge $\mathbf{A}_{m,n}$ represents the weighted directed relation from vertex v_m to v_n , i.e.

$$\mathbf{A}_{m,n} = \{a_{mn}\}_{m,n=1}^N \in \mathbb{R}^{N \times N},$$

where a_{mn} is equal to 1 if m is connected to n , 0 otherwise. In the case of an undirected graph, where the relation is bidirectional, we have $\mathbf{A}_{n,m} = \mathbf{A}_{m,n}$, resulting in the symmetry of the matrix $\mathbf{A}$ [2, 4].

The graph signal is defined as a mapping that maps the vertex v_n to a signal coefficient $x_n \in \mathbb{C}$, which can be represented as a vector $\mathbf{x} = [x_0, x_1, \dots, x_{N-1}]^T$. The GFT utilizes the eigenvectors of the adjacency matrix $\mathbf{A}$. For an orthogonalizable and eigendecomposable adjacency matrix [2]

$$\mathbf{A} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^{-1} = \sum_{i=0}^{N-1} \lambda_{\mathbf{A}_i} u_i u_i^*, \quad (1)$$

where the eigenvalues in the corresponding diagonal matrix $\mathbf{\Lambda}_{\mathbf{A}} \in \mathbb{C}^{N \times N}$ are $\lambda_0(\mathbf{A}), \dots, \lambda_{N-1}(\mathbf{A})$, and the columns $u_0, u_1, \dots, u_{N-1}$ of $\mathbf{V}$ are the eigenvectors of $\mathbf{A}$. The eigenvalues are the graph frequencies in descending order from low to high frequency [4].

Definition 1. The GFT of $\mathbf{x} \in \mathbb{C}^N$ is defined as

$$\hat{\mathbf{x}} = \mathbf{V}^{-1} \mathbf{x}, \quad (2)$$

and the inverse GFT (IGFT) is

$$\mathbf{x} = \mathbf{V} \hat{\mathbf{x}}, \quad (3)$$

where $\mathbf{V}^{-1}$ is the GFT matrix. As our framework relies on GLCT, we give the definition of GLCT in the following.

2.2 Graph Linear Canonical Transform

Parallel to the eigenvalue decomposition method of defining discrete linear canonical transform [18], we proposed in [11] the definition of GLCT.

Definition 2. The $\mathbf{M}$ matrix GLCT of $\mathbf{x} \in \mathbb{C}^N$ is defined as

$$\hat{\mathbf{x}} = \mathbf{O}^M \mathbf{x} = \left\{ \mathbf{\Lambda}^\xi \mathbf{Q}_\beta \mathbf{\Lambda}^\alpha \mathbf{Q}^\top \right\} \mathbf{x}, \quad (4)$$

and the inverse GLCT is

$$\mathbf{x} = (\mathbf{O}^M)^{-1} \hat{\mathbf{x}} = \left\{ \mathbf{Q} \mathbf{\Lambda}^{-\alpha} \mathbf{Q}_\beta^\top \mathbf{\Lambda}^{-\xi} \right\} \hat{\mathbf{x}}, \quad (5)$$

where the matrix $\mathbf{M}$ entries are (a, b, c, d) , which is called a GLCT parameter matrix, and $ad - bc = 1$, $a, b, c, d \in \mathbb{R}$, and the parameter relations between $\mathbf{M}$ and (ξ, β, α) are

$$\begin{aligned} \xi &= \frac{ac + bd}{a^2 + b^2}, \quad \beta = \sqrt{a^2 + b^2}, \\ \alpha &= \arccos\left(\frac{a}{\beta}\right) = \arcsin\left(\frac{b}{\beta}\right). \end{aligned} \quad (6)$$

The Λ^ξ matrix, $\mathbf{Q}_\beta$ matrix, and Λ^α matrix are the matrices in the graph chirp multiplication (CM), graph scale transform, and graph fractional Fourier transform (GFRFT), respectively [11]. And $\mathbf{Q}$ and Λ are obtained through the eigendecomposition of $\mathbf{V}^{-1} = \mathbf{Q}\Lambda\mathbf{Q}^{-1}$.

Remark 1. It is easy to verify that when $a = \cos \alpha$, $b = \sin \alpha$, $c = -\sin \alpha$, and $d = \cos \alpha$, the GLCT operator will be the GFRFT operator [9], when $a = 0$, $b = 1$, $c = -1$, and $d = 0$, the GLCT operator will be the GFT operator.

2.3 Sampling in the GLCT Domain

Sampling is one of the fundamental research areas of GLCT. The essence of sampling is to recover the entire set of information from a small subset. For an M -bandlimited signal [13], we have following definition for the M -bandlimited set.

Definition 3. The closed subspace of graph signals in $\mathbb{C}^N$ with a bandwidth of at most $|\mathcal{F}|$ is denoted as $\text{BL}_{|\mathcal{F}|}(\mathbf{O}^M)$, where $\mathbf{O}^M$ is defined as in Eq. (4).

For a subset of vertices $\mathcal{S} \subseteq \mathcal{V}$, the vertex-limiting operator is defined as follows

$$\mathbf{D}_{\mathcal{S}} = \text{diag}(d_1, \dots, d_i, \dots, d_N), \quad (7)$$

where, $d_i = 1$, if $i \in \mathcal{S}$, and $d_i = 0$ if $i \notin \mathcal{S}$, $i = 1, \dots, N$.

Similarly, the index subset $\mathcal{F} \subseteq \hat{\mathcal{G}}$, where $\hat{\mathcal{G}} = \{1, \dots, N\}$ represents the set of all graph spectral indices, the spectral-limiting operator as

$$\mathbf{B}_{\mathcal{F}}^M = (\mathbf{O}^M)^{-1} \Sigma_{\mathcal{F}} \mathbf{O}^M, \quad (8)$$

where, $\Sigma_{\mathcal{F}}$ is a diagonal matrix whose definition is consistent with $\mathbf{D}_{\mathcal{S}}$. Obviously, the operators $\mathbf{D}_{\mathcal{S}}$ and $\mathbf{B}_{\mathcal{F}}^M$ are symmetric and positive semi-definite, acting as self-adjoint operators. We use $\mathbf{D}$ and $\mathbf{B}^M$ to simplify these two operators, respectively. Additionally, for a perfectly localized signal $\mathbf{x}$, it holds that $\mathbf{D}\mathbf{x} = \mathbf{x}$ and $\mathbf{B}^M\mathbf{x} = \mathbf{x}$ [15].

The interpolation (recovery) operator $\mathbf{R} \in \mathbb{C}^{N \times |\mathcal{S}|}$ is characterized by the following conditions:

- (1) $\mathbf{R}$ spans the space $\text{BL}_{|\mathcal{F}|}(\mathbf{O}^M)$;
- (2) $\mathbf{R}\mathbf{D}$ functions as a projection operator, achieving perfect recovery

$$\mathbf{x} = \mathbf{R}\mathbf{D}\mathbf{x} = \mathbf{R}\mathbf{x}_{\mathcal{S}}, \quad \text{for all } \mathbf{x} \in \text{BL}_{|\mathcal{F}|}(\mathbf{O}^M), \quad (9)$$

where $\mathbf{x}_{\mathcal{S}}$ denotes the sampled graph signal.

If the original signal $\mathbf{x}$ can be perfectly recovered, then

$$\mathbf{x} - \mathbf{R}\mathbf{D}\mathbf{x} = \mathbf{x} - \mathbf{R}(\mathbf{I} - \bar{\mathbf{D}})\mathbf{x} = \mathbf{x} - \mathbf{R}(\mathbf{I} - \bar{\mathbf{D}}\mathbf{B}^M)\mathbf{x},$$

where $\bar{\mathbf{D}}$ is the complement of $\mathbf{D}$. The perfect recoverability of $\mathbf{x}$ implies the existence of $\mathbf{R}$, which is tantamount to the invertibility of $(\mathbf{I} - \bar{\mathbf{D}}\mathbf{B}^M)$. Furthermore, since $\mathbf{x} \in \text{BL}_{|\mathcal{F}|}(\mathbf{O}^M)$,

$$(\mathbf{I} - \bar{\mathbf{D}}\mathbf{B}^M)\mathbf{x} = (\mathbf{I} - \bar{\mathbf{D}})\mathbf{x} = \mathbf{D}\mathbf{x} = \mathbf{D}\mathbf{B}^M\mathbf{x}.$$

We can get the conclusion that $\mathbf{R} = (\mathbf{D}\mathbf{B}^M)^{-1}$. Unfortunately, the rank of $\mathbf{D}\mathbf{B}^M$ may be less than full. In this situation, we use the pseudo-inverse to define $\mathbf{R} = (\mathbf{D}\mathbf{B}^M)^\dagger$.

3 FAST SAMPLING SET SELECTION IN GLCT DOMAIN

This section introduces a proposed method for fast sampling set selection in the GLCT domain. Firstly, we present a novel reconstruction approach for missing graph signals based on the filtering

operators in the GLCT domain. Subsequently, we select sampling vertices to minimize the reconstruction error. Finally, a fast sampling set selection method is proposed using the localization filter operator [17].

3.1 A New Reconstruction Method in GLCT Domain

Before proposing the reconstruction method, we need to define a polynomial filter. Let $h(\lambda_i(\mathbf{A}))$ be the graph kernel, where $\lambda_i(\mathbf{A})$ is the eigenvalue of the adjacency matrix $\mathbf{A}$. Therefore, the filter in the GLCT domain is given by

$$\mathbf{x} = (\mathbf{O}^M)^{-1} h(\Lambda_{\mathbf{A}}) \mathbf{O}^M \mathbf{y},$$

where $\mathbf{x}$ and $\mathbf{y}$ represent the output and input signals, respectively, and $\Lambda_{\mathbf{A}} = \text{diag}(\lambda_0(\mathbf{A}), \dots, \lambda_{N-1}(\mathbf{A}))$.

The localized filter operator [19] is defined as follows

$$T_{h,i}^M(n) = \sqrt{N} \sum_{l=0}^{N-1} h(\lambda_l) u_l^*(i) u_l(n), \quad (10)$$

where n represents the element of the operator, i represents the center vertex. $h(\lambda_l)$ can be a low-pass filter, and u_l is the l th column of matrix $(\mathbf{O}^M)^{-1}$. The operators are combined into the following matrix form

$$\mathbf{T}^M = \begin{bmatrix} T_{h,0}^M & T_{h,1}^M & \dots & T_{h,N-1}^M \end{bmatrix} = \sqrt{N} (\mathbf{O}^M)^{-1} h(\Lambda_{\mathbf{A}}) \mathbf{O}^M. \quad (11)$$

The missing values are reconstructed through a weighted linear combination using $\mathbf{T}^M$. Therefore, we elucidate the proposed reconstruction method in the following theorem.

Theorem 1. For a signal $\mathbf{x} \in \text{BL}_{|\mathcal{F}|}(\mathbf{O}^M)$, where the graph spectral bandwidth is $|\mathcal{F}|$ and the vertex domain bandwidth is $|\mathcal{S}| \geq |\mathcal{F}|$, if the sampled signal $\mathbf{x}_{\mathcal{S}}$ can be perfectly recovered, then the following holds

$$\mathbf{x}_{\mathcal{R}} = \mathbf{T}_{\mathcal{V}\mathcal{S}}^M (\mathbf{T}_{\mathcal{S}}^M)^\dagger \mathbf{x}_{\mathcal{S}}, \quad (12)$$

where, $\mathbf{x}_{\mathcal{R}}$ denotes the recovery signal, while $\mathcal{V}$ and $\mathcal{S}$ represent the sets of all vertices and sampled vertices, respectively, with $\mathcal{S} \subseteq \mathcal{V}$.

PROOF. $\mathbf{x}_{\mathcal{R}}$ can be rewritten as

$$\begin{aligned} \mathbf{x}_{\mathcal{R}} &= \mathbf{T}_{\mathcal{V}\mathcal{S}}^M (\mathbf{T}_{\mathcal{S}}^M)^\dagger \mathbf{x}_{\mathcal{S}} \\ &= \left[(\mathbf{O}^M)^{-1} h(\Lambda_{\mathbf{A}}) \mathbf{O}^M \right]_{\mathcal{V}\mathcal{S}} \left(\left[(\mathbf{O}^M)^{-1} h(\Lambda_{\mathbf{A}}) \mathbf{O}^M \right]_{\mathcal{S}} \right)^\dagger \mathbf{x}_{\mathcal{S}} \\ &= (\mathbf{O}^M)^{-1} h(\Lambda_{\mathbf{A}}) \mathbf{O}_{\mathcal{S}\mathcal{V}}^M \left((\mathbf{O}^M)^{-1}_{\mathcal{S}\mathcal{V}} h(\Lambda_{\mathbf{A}}) \mathbf{O}_{\mathcal{S}\mathcal{V}}^M \right)^\dagger \mathbf{x}_{\mathcal{S}} \\ &= (\mathbf{O}^M)^{-1} h^{\frac{1}{2}}(\Lambda_{\mathbf{A}}) \left((\mathbf{O}^M)^{-1}_{\mathcal{S}\mathcal{V}} h^{\frac{1}{2}}(\Lambda_{\mathbf{A}}) \right)^\dagger \mathbf{x}_{\mathcal{S}}, \end{aligned}$$

The above expression can also be written as

$$\mathbf{x}_{\mathcal{R}} = (\mathbf{O}^M)^{-1} \left(\frac{h(\Lambda_{\mathbf{A}})}{\rho} \right)^{\frac{1}{2}} \left((\mathbf{O}^M)^{-1}_{\mathcal{S}\mathcal{V}} \left(\frac{h(\Lambda_{\mathbf{A}})}{\rho} \right)^{\frac{1}{2}} \right)^\dagger \mathbf{x}_{\mathcal{S}}, \quad (13)$$

where $\rho = \min_{0 \leq i \leq |\mathcal{F}|-1} h(\lambda_i(\mathbf{A}))$. It satisfies that when $i \geq |\mathcal{F}|$, $(h(\Lambda_{\mathbf{A}})/\rho)^{1/2}$ tends to 0. Therefore, we can obtain the following approximation

$$\left((\mathbf{O}^M)^{-1}_{\mathcal{S}\mathcal{V}} \left(\frac{h(\Lambda_{\mathbf{A}})}{\rho} \right)^{\frac{1}{2}} \right)^\dagger \approx \left(\frac{h(\Lambda_{\mathbf{A}})}{\rho} \right)^{-\frac{1}{2}} \cdot \left((\mathbf{O}^M)^{-1}_{\mathcal{S}\mathcal{F}} \right)^\dagger. \quad (14)$$

Table 2: Minimization Methods of Error Covariance Matrix

Optimal Methods	Objectives
A-optimal	$\min_{S \subseteq \mathcal{V}} \text{tr}[(\mathbf{T}_S)^{-1}]$
D-optimal	$\min_{S \subseteq \mathcal{V}} \det[(\mathbf{T}_S)^{-1}]$
E-optimal	$\min_{S \subseteq \mathcal{V}} \ (\mathbf{T}_{S^c}^{1/2})^\dagger\ _2$
T-optimal	$\max_{S \subseteq \mathcal{V}} \text{tr}[(\mathbf{T}_S)]$

Through (13) and (14), it can be rewritten as

$$\begin{aligned} \mathbf{x}_R &= (\mathbf{O}^M)_{\mathcal{V}^c}^{-1} \left((\mathbf{O}^M)_{\mathcal{S}}^{-1} \right)^\dagger \mathbf{x}_S = (\mathbf{O}^M)_{\mathcal{V}}^{-1} \Sigma_{\mathcal{F}} \mathbf{O}_{\mathcal{S}^c \mathcal{V}}^M \mathbf{x}_S \\ &= \mathbf{D}_S (\mathbf{O}^M)_{\mathcal{V}}^{-1} \Sigma_{\mathcal{F}} \mathbf{O}_{\mathcal{V}}^M \mathbf{x}_S = \mathbf{D} \mathbf{B}^M \mathbf{x}_S = \mathbf{R} \mathbf{x}_S, \end{aligned} \quad (15)$$

where $\mathbf{D}_S$ and $\Sigma_{\mathcal{F}}$ are defined in (7) and (8), respectively. $\square$

Hence, according to equations (9), (12), and (15), we have $\mathbf{R} \approx \mathbf{T}_{\mathcal{V}^c \mathcal{S}}^M (\mathbf{T}_S^M)^\dagger$. Moving forward, our focus will be solely on understanding the matrix $\mathbf{T}$.

3.2 Reconstruction Error

When the sampled signal includes Gaussian random noise, we consider the reconstruction error through (12), where the sampled signal $\mathbf{x}_S = \mathbf{D} \mathbf{x} + \mathbf{n}$. The recovered signal $\mathbf{x}_R$ is then given by $\mathbf{x}_R = \mathbf{R} \mathbf{x}_S = \mathbf{R} \mathbf{D} \mathbf{x} + \mathbf{R} \mathbf{n} = \mathbf{x} + \mathbf{R} \mathbf{n}$. Therefore, the error $\mathbf{e}$ is represented as

$$\mathbf{e} = \mathbf{x}_R - \mathbf{x} = \mathbf{R} \mathbf{n} = \mathbf{T}_{\mathcal{V}^c \mathcal{S}}^M (\mathbf{T}_S^M)^\dagger \mathbf{n} \quad (16)$$

The above equation can be expanded and simplified to

$$\begin{aligned} \mathbf{e} &= (\mathbf{O}^M)^{-1} h(\Lambda_A) \mathbf{O}_{\mathcal{S}^c \mathcal{V}}^M \left((\mathbf{O}^M)_{\mathcal{S}^c \mathcal{V}}^{-1} h(\Lambda_A) \mathbf{O}_{\mathcal{S}^c \mathcal{V}}^M \right)^\dagger \mathbf{n} \\ &= (\mathbf{O}^M)^{-1} h^{\frac{1}{2}}(\Lambda_A) \mathbf{O}^M (\mathbf{O}^M)^{-1} h^{\frac{1}{2}}(\Lambda_A) \mathbf{O}_{\mathcal{S}^c \mathcal{V}}^M \\ &\quad \times \left((\mathbf{O}^M)_{\mathcal{S}^c \mathcal{V}}^{-1} h^{\frac{1}{2}}(\Lambda_A) \mathbf{O}^M (\mathbf{O}^M)^{-1} h^{\frac{1}{2}}(\Lambda_A) \mathbf{O}_{\mathcal{S}^c \mathcal{V}}^M \right)^\dagger \mathbf{n} \\ &= (\mathbf{T}^M)^{1/2} \left((\mathbf{T}^M)_{\mathcal{S}^c \mathcal{V}}^{1/2} \right)^\dagger \mathbf{n} \end{aligned}$$

Then, calculate the error covariance matrix, i.e.,

$$\begin{aligned} \mathbf{E} &= \mathbb{E}(\mathbf{e} \mathbf{e}^*) \\ &= (\mathbf{T}^M)^{1/2} \left((\mathbf{T}^M)_{\mathcal{S}^c \mathcal{V}}^{1/2} \right)^\dagger \mathbf{I}_S \left((\mathbf{T}^M)_{\mathcal{S}^c \mathcal{V}}^{-1/2} \right)^\dagger (\mathbf{T}^M)^{1/2} \\ &= (\mathbf{T}^M)^{1/2} \left((\mathbf{T}^M)_{\mathcal{S}^c \mathcal{V}}^{-1/2} (\mathbf{T}^M)_{\mathcal{S}^c \mathcal{V}}^{1/2} \right)^\dagger (\mathbf{T}^M)^{1/2} \end{aligned}$$

To minimize the error, we should optimize the strategy by minimizing or maximizing the trace, determinant, or maximum eigenvalue of $\mathbf{E}$. Table 2 summarizes these methods [13].

3.3 Fast Sampling Set Selection Method Based on Localization Filter Operators

While the cost functions based on Table 2 can effectively address the optimal selection problem of sampling sets in the GLCT domain, maximizing or minimizing these cost functions requires eigendecomposition, resulting in high computational complexity. Taking inspiration from [17], we leverage localization filter operators $\mathbf{T}_{h,i}^M$ to reduce computational complexity and minimize time costs.

The missing samples are interpolated by $\mathbf{T}^M$ located at all observed samples. Therefore, given the “support” of $\mathbf{T}_{h,i}^M$, which can be seen as the region where the i th vertex can interpolate unobserved samples, we aim to maximize the “coverage area” of the sampling set $\mathcal{S}$, represented as $\phi(|\mathbf{T}_{h,i}^M|)$, i.e., maximizing $\sum_{i \in \mathcal{S}} \phi(|\mathbf{T}_{h,i}^M|)$. Our objective is to select local operators that satisfy the following conditions: each operator $\mathbf{T}_{h,i}^M$ has large energy in the first sense, and for vertices i and j ($i \neq j$), the overlap region of their local filter operators should be small. Therefore, we can obtain

$$\mathcal{S}^{\text{opt}} = \arg \max_{S \subseteq \mathcal{V}} \sum_{i \in S} \left\langle \left(\epsilon \mathbf{1}_{N \times 1} - \sum_{j \in S, j \neq i} |\mathbf{T}_{h,j}^M| \right), |\mathbf{T}_{h,i}^M| \right\rangle, \quad (17)$$

where, ϵ is a real-valued constant, and $\epsilon = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \sum_{j \in S} |\mathbf{T}_{h,j}^M(i)|$ can be obtained from experiments. Similar to [2], a greedy algorithm is employed to optimize the cost function. The algorithm adds a vertex y^{opt} in the m th iteration by selecting vertices that satisfy the following function

$$y^{\text{opt}} = \arg \max_{y \in \mathcal{V} \setminus S} \left\langle R \left(\epsilon \mathbf{1}_{N \times 1} - \sum_{j \in S_m} |\mathbf{T}_{h,j}^M| \right), |\mathbf{T}_{h,y}^M| \right\rangle, \quad (18)$$

where, $R(\cdot)$ is defined such that $[R(x)](i) = x(i)$ if $x(i) \geq 0$, and 0 otherwise.

In computing the weighted norm of $\mathbf{T}_{h,y}^M$ in (18), if the i th vertex has already been covered, a smaller weight is assigned to $\mathbf{T}_{h,y}^M(i)$. Additionally, for $j \in S_m$, the weight of $|\mathbf{T}_{h,j}^M|$ is relatively large at the i th vertex. This is due to the fact that the kernel $h(\lambda_{A_i})$ is a polynomial function, and $\mathbf{T}^M = \sqrt{N} h(\mathbf{A}^M)$, where $\mathbf{A}^M = (\mathbf{O}^M)^{-1} \Lambda_A \mathbf{O}^M$ [13]. Consequently, there is no need to perform eigendecomposition on the adjacency matrix in the GLCT domain to calculate (18).

Table 3 compares the computational complexity of the GLCT domain sampling algorithm based on prior research [13] with the fast sampling algorithm proposed in this section. Here, T_M represents the convergence iteration count for the M th eigenpair, C is a constant, and J is the number of non-zero elements in $\mathbf{T}^M$. As shown in the table, the computational complexity of the localization operator is significantly lower than the complexity of computing eigenpairs in other methods.

4 THE RELATIONSHIP BETWEEN FAST SAMPLING ALGORITHMS AND PREVIOUS ALGORITHMS

Previously proposed objective functions in the GLCT domain [13] can be rephrased using localization filter operators with specific kernels. Table 4 summarizes them. We illustrate the potential connections between existing methods based on GLCT and the proposed approach.

4.1 GLCT Sampling Based on Error Minimization

Assuming $|\mathcal{F}| \leq |\mathcal{S}|$ and employing (12) for recovery. There are two objective functions based on minimizing the error to select the optimal sampling set.

Table 3: Computational Complexity of Sampling Methods in the GLCT Domain

	MinFro	MinPinv	MaxSig	MaxSigMin	MaxVol	Proposed Method
Operator Computations	$\mathcal{O}((\mathcal{E} M + CM^3)T_M)$					$\mathcal{O}((\mathcal{E} + N) + J)$
Sampling Set Selection	$\mathcal{O}(NM)$	$\mathcal{O}(NM^4)$	$\mathcal{O}(NM^4)$	$\mathcal{O}(NM^4)$	$\mathcal{O}(M^3)$	$\mathcal{O}(JM)$

Table 4: The Speed up of Proposed Method Relative to The Previous Method

Methods	Objective Functions	Localized Filter Operators
MinFro	$\arg \min_{S \subseteq \mathcal{V}} \left\ \left(\Sigma_{\mathcal{F}} \mathbf{O}^{\mathbf{M}} \mathbf{D}_S \right)^{\dagger} \right\ _F$	$\arg \min_{S \subseteq \mathcal{V}} \text{tr} \left[\left(\mathbf{T}_S^{\mathbf{M}} \right)^{-1} \right]$
MinPinv	$\arg \min_{S \subseteq \mathcal{V}} \sum_{i=1}^{ \mathcal{F} } \frac{1}{\sigma_i(\Sigma_{\mathcal{F}} \mathbf{O}^{\mathbf{M}} \mathbf{D}_S)}$	$\arg \min_{S \subseteq \mathcal{V}} \text{tr} \left[\left(\mathbf{T}_S^{\mathbf{M}} \right)^{-1} \right]$
MaxSig	$\arg \max_{S \subseteq \mathcal{V}} \left\ \Sigma_{\mathcal{F}} \mathbf{O}^{\mathbf{M}} \mathbf{D}_S \right\ _F$	$\arg \max_{S \subseteq \mathcal{V}} \text{tr} \left[\mathbf{T}_S^{\mathbf{M}} \right]$
MaxSigMin	$\arg \max_{S \subseteq \mathcal{V}} \sigma_{\min} \left(\mathbf{O}_{S\mathcal{F}}^{-\mathbf{M}} \right)$	$\arg \min_{S \subseteq \mathcal{V}} \left\ \left(\mathbf{T}_{\mathcal{V}S}^{\mathbf{M}} \right)^{\dagger} \right\ _2$
MaxVol	$\arg \max_{S \subseteq \mathcal{V}} \det \left[\mathbf{O}_{S\mathcal{F}}^{-\mathbf{M}} \mathbf{O}_{S\mathcal{F}}^{\mathbf{M}} \right]$	$\arg \max_{S \subseteq \mathcal{V}} \det \left[\mathbf{T}_S^{\mathbf{M}} \right]$

Then, the recovery error is $e = (\mathbf{O}^{\mathbf{M}})^{-1}_{\mathcal{V}\mathcal{F}} \left((\mathbf{O}^{\mathbf{M}})^{-1}_{S\mathcal{F}} \right)^{\dagger} n$.

MaxSigMin: When adopting the 2-norm to minimize the error, there is

$$S^{\text{opt}} = \arg \min_{S \subseteq \mathcal{V}} \|e\|_2 = \arg \min_{S \subseteq \mathcal{V}} \left\| (\mathbf{O}^{\mathbf{M}})^{-1}_{\mathcal{V}\mathcal{F}} \left((\mathbf{O}^{\mathbf{M}})^{-1}_{S\mathcal{F}} \right)^{\dagger} n \right\|_2.$$

The objective function can be further written through the Cauchy-Schwarz inequality as

$$\left\| (\mathbf{O}^{\mathbf{M}})^{-1}_{\mathcal{V}\mathcal{F}} \left((\mathbf{O}^{\mathbf{M}})^{-1}_{S\mathcal{F}} \right)^{\dagger} n \right\|_2 \leq \left\| (\mathbf{O}^{\mathbf{M}})^{-1}_{\mathcal{V}\mathcal{F}} \right\|_2 \left\| \left((\mathbf{O}^{\mathbf{M}})^{-1}_{S\mathcal{F}} \right)^{\dagger} \right\|_2 \|n\|_2.$$

Since $\left\| \left((\mathbf{O}^{\mathbf{M}})^{-1}_{S\mathcal{F}} \right)^{\dagger} \right\|_2$ is the only variable under consideration, with $\left\| (\mathbf{O}^{\mathbf{M}})^{-1}_{\mathcal{V}\mathcal{F}} \right\|_2$ and $\|n\|_2$ being fixed, the goal is to maximize the minimum singular value of $(\mathbf{O}^{\mathbf{M}})^{-1}_{S\mathcal{F}}$. Therefore, it is equivalent to maximize the minimum singular value of $(\mathbf{O}^{\mathbf{M}})^{-1}_{S\mathcal{F}}$.

$$\begin{aligned} S^{\text{opt}} &= \arg \max_{S \subseteq \mathcal{V}} \sigma_{\min} \left((\mathbf{O}^{\mathbf{M}})^{-1}_{S\mathcal{F}} \right) \\ &= \arg \max_{S \subseteq \mathcal{V}} \sigma_{\min} \left(\mathbf{O}_{S\mathcal{F}}^{\mathbf{M}} \right) \\ &= \arg \max_{S \subseteq \mathcal{V}} \sigma_{\min} \left(\Sigma_{\mathcal{F}} \mathbf{O}^{\mathbf{M}} \mathbf{D}_S \right) \\ &= \arg \max_{S \subseteq \mathcal{V}} \sigma_{\min} \left((\mathbf{O}^{\mathbf{M}})^{-1} \Sigma_{\mathcal{F}} \mathbf{O}^{\mathbf{M}} \mathbf{D}_S \right) \\ &= \arg \max_{S \subseteq \mathcal{V}} \sigma_{\min} \left(\mathbf{T}_{\mathcal{V}S}^{\mathbf{M}} \right) \\ &= \arg \min_{S \subseteq \mathcal{V}} \left\| \left(\mathbf{T}_{\mathcal{V}S}^{\mathbf{M}} \right)^{\dagger} \right\|_2, \end{aligned} \quad (19)$$

where $\mathbf{T}^{\mathbf{M}} = (\mathbf{O}^{\mathbf{M}})^{-1} \Sigma_{\mathcal{F}} \mathbf{O}^{\mathbf{M}}$, and $h(\Lambda_A) = \Sigma_{\mathcal{F}}$ is the ideal low-pass filter. Additionally, $\mathbf{T}^{\mathbf{M}} = \mathbf{B}^{\mathbf{M}}$ represents the localization operator in the spectral domain.

This optimization problem uses a greedy algorithm to select vertex y^{opt} at the m th iteration

$$y^{\text{opt}} = \arg \min_{y \in \mathcal{V} \setminus S_m} \left\| \left(\mathbf{T}_{\mathcal{V}(S_m \cup \{y\})}^{\mathbf{M}} \right)^{\dagger} \right\|_2. \quad (20)$$

MinTrac: Minimizing the trace of the error covariance matrix $\mathbf{E}$, there is

$$\begin{aligned} S^{\text{opt}} &= \arg \min_{S \subseteq \mathcal{V}} \text{tr} [\mathbf{E}] = \arg \min_{S \subseteq \mathcal{V}} \text{tr} [ee^*] \\ &= \arg \min_{S \subseteq \mathcal{V}} \text{tr} \left[(\mathbf{O}^{\mathbf{M}})^{-1}_{\mathcal{V}\mathcal{F}} \left(\mathbf{O}_{S\mathcal{F}}^{\mathbf{M}} (\mathbf{O}^{\mathbf{M}})^{-1}_{S\mathcal{F}} \right)^{-1} (\mathbf{O}^{\mathbf{M}})^{-1}_{\mathcal{V}\mathcal{F}} \right] \\ &= \arg \min_{S \subseteq \mathcal{V}} \text{tr} \left[\left(\mathbf{O}_{S\mathcal{F}}^{\mathbf{M}} (\mathbf{O}^{\mathbf{M}})^{-1}_{S\mathcal{F}} \right)^{-1} \right] \end{aligned}$$

We express the above equation in terms of the matrix $\mathbf{T}^{\mathbf{M}}$ as follows

$$\begin{aligned} S^{\text{opt}} &= \arg \min_{S \subseteq \mathcal{V}} \text{tr} \left[\left((\mathbf{O}^{\mathbf{M}})^{-1}_{S\mathcal{F}} \mathbf{O}_{S\mathcal{F}}^{\mathbf{M}} \right)^{-1} \right] \\ &= \arg \min_{S \subseteq \mathcal{V}} \text{tr} \left[\left(\left[(\mathbf{O}^{\mathbf{M}})^{-1} \Sigma_{\mathcal{F}} \mathbf{O}^{\mathbf{M}} \right]_S \right)^{-1} \right] \\ &= \arg \min_{S \subseteq \mathcal{V}} \text{tr} \left[\left(\mathbf{T}_S^{\mathbf{M}} \right)^{-1} \right]. \end{aligned} \quad (21)$$

This optimization problem also uses a greedy algorithm to select vertex y^{opt} at the m th iteration

$$y^{\text{opt}} = \arg \min_{y \in \mathcal{V} \setminus S_m} \text{tr} \left[\left(\mathbf{T}_{(S_m \cup \{y\})}^{\mathbf{M}} \right)^{-1} \right]. \quad (22)$$

4.2 GLCT Sampling Based on Localized Basis

Similarly, assuming $|\mathcal{F}| \leq |\mathcal{S}|$ and using the vertex localization operator $\mathbf{D}$ and the spectral domain localization operator $\mathbf{B}^{\mathbf{M}}$ to recover the original signal $\mathbf{x}$. If $\mathbf{x} \in \text{BL}_{|\mathcal{F}|}(\mathbf{O}^{\mathbf{M}})$, this method can perfectly recover the original signal. The method has three objective functions, and the recovery error is $e = \left(\mathbf{B}^{\mathbf{M}} \mathbf{D} \mathbf{B}^{\mathbf{M}} \right)^{\dagger} \mathbf{D} n$.

MinPinv: Minimize the Frobenius norm of matrix $(\mathbf{B}^M \mathbf{D} \mathbf{B}^M)^\dagger$

$$\mathbf{S}^{\text{opt}} = \arg \min_{\mathbf{S} \subseteq \mathcal{V}} \left\| (\mathbf{B}^M \mathbf{D} \mathbf{B}^M)^\dagger \mathbf{D} \mathbf{n} \right\|_F$$

Its objective function also can be further written through the Cauchy-Schwarz inequality as

$$\left\| (\mathbf{B}^M \mathbf{D} \mathbf{B}^M)^\dagger \mathbf{D} \mathbf{n} \right\|_F \leq \left\| (\Sigma_{\mathcal{F}} \mathbf{O}^M \mathbf{D}_S)^\dagger \right\|_F \|\mathbf{D} \mathbf{n}\|_F.$$

Since $\|\mathbf{D} \mathbf{n}\|_F$ is fixed, we only need to consider $\left\| (\Sigma_{\mathcal{F}} \mathbf{O}^M \mathbf{D}_S)^\dagger \right\|_F$.

$$\begin{aligned} \mathbf{S}^{\text{opt}} &= \arg \min_{\mathbf{S} \subseteq \mathcal{V}} \left\| (\Sigma_{\mathcal{F}} \mathbf{O}^M \mathbf{D}_S)^\dagger \right\|_F \\ &= \arg \min_{\mathbf{S} \subseteq \mathcal{V}} \sum_{i=1}^{|\mathcal{F}|} \frac{1}{\sigma_i(\Sigma_{\mathcal{F}} \mathbf{O}^M \mathbf{D}_S)} \\ &= \arg \min_{\mathbf{S} \subseteq \mathcal{V}} \sum_{i=1}^{|\mathcal{F}|} \frac{1}{\sigma_i((\mathbf{O}^M)^{-1} \Sigma_{\mathcal{F}} \mathbf{O}^M \mathbf{D}_S)} \\ &= \arg \min_{\mathbf{S} \subseteq \mathcal{V}} \sum_{i=1}^{|\mathcal{F}|} \frac{1}{\sigma_i(\mathbf{T}_{\mathcal{V}S}^M)} \\ &= \arg \min_{\mathbf{S} \subseteq \mathcal{V}} \left\| (\mathbf{T}_{\mathcal{V}S}^M)^\dagger \right\|_F \\ &= \arg \min_{\mathbf{S} \subseteq \mathcal{V}} \text{tr} \left[(\mathbf{T}_S^M)^{-1} \right]. \end{aligned} \quad (23)$$

For the matrix $\mathbf{T}_S^M$, we use a greedy algorithm to select vertex y^{opt} at the m th iteration

$$y^{\text{opt}} = \arg \min_{y \in \mathcal{V} \setminus S_m} \text{tr} \left[(\mathbf{T}_{(S_m \cup \{y\})}^M)^{-1} \right], \quad (24)$$

The optimization algorithm for this method is consistent with the *MinTrac* based on error minimization.

MaxSig: Similar to *MinPinv*, maximizing the Frobenius norm of $\mathbf{B}^M \mathbf{D} \mathbf{B}^M$ is sufficient.

$$\begin{aligned} \mathbf{S}^{\text{opt}} &= \arg \max_{\mathbf{S} \subseteq \mathcal{V}} \left\| \Sigma_{\mathcal{F}} \mathbf{O}^M \mathbf{D}_S \right\|_F \\ &= \arg \max_{\mathbf{S} \subseteq \mathcal{V}} \left\| \mathbf{T}_{\mathcal{V}S}^M \right\|_F \\ &= \arg \max_{\mathbf{S} \subseteq \mathcal{V}} \text{tr} \left[\mathbf{T}_S^M \right]. \end{aligned} \quad (25)$$

This method can select the $|\mathbf{S}|$ column of $\mathbf{T}_S^M$ with the maximum 2 norm to solve, using a greedy algorithm to select vertex y^{opt} at the m th iteration

$$y^{\text{opt}} = \arg \max_{y \in \mathcal{V} \setminus S_m} \text{tr} \left[\mathbf{T}_{(S_m \cup \{y\})}^M \right]. \quad (26)$$

MaxVol: By calculating the determinant of $(\mathbf{O}^M)_{S\mathcal{F}}^{-1} \mathbf{O}_{S\mathcal{F}}^M$, the objective is to maximize the volume formed by the parallelepiped

defined by $\mathbf{O}_{S\mathcal{F}}^M$.

$$\begin{aligned} \mathbf{S}^{\text{opt}} &= \arg \max_{\mathbf{S} \subseteq \mathcal{V}} \det \left[(\mathbf{O}^M)_{S\mathcal{F}}^{-1} \mathbf{O}_{S\mathcal{F}}^M \right] \\ &= \arg \max_{\mathbf{S} \subseteq \mathcal{V}} \det \left[((\mathbf{O}^M)^{-1} \Sigma_{\mathcal{F}} \mathbf{O}^M)_S \right] \\ &= \arg \max_{\mathbf{S} \subseteq \mathcal{V}} \det \left[\mathbf{T}_S^M \right]. \end{aligned} \quad (27)$$

This optimization problem also uses a greedy algorithm to select vertex y^{opt} at the m th iteration

$$y^{\text{opt}} = \arg \max_{y \in \mathcal{V} \setminus S_m} \det \left[\mathbf{T}_{(S_m \cup \{y\})}^M \right]. \quad (28)$$

In the case of using the ideal kernel filtering for localization, the prior GLCT domain sampling set selection methods shown in (19)-(28) can be related to the localization filtering operator $\mathbf{T}^M$, and the sampling methods can vary with different kernels [17]. Table 4 summarizes the objective functions represented by the localization operators in the GLCT domain.

In fact, all existing GLCT domain sampling set selection methods introduced in this paper can be considered as special cases of the proposed error-based sampling set selection, aiming to minimize the reconstruction error.

5 NUMERICAL EXPERIMENTS

In the numerical experiments, we employed the following five types of graphs: a random sensor graph, a randomly generated community graph, the Minnesota traffic graph, and the Swiss roll graph. For the comparison of execution time, we selected a randomly generated community graph with varying numbers of vertices. To compare recovery errors, we utilized the Swiss roll graph and the Roman traffic flow graph. All experiments were conducted in MATLAB R2023a using the graph signal processing toolbox (GSPBox) in MATLAB [20].

5.1 Comparison of Selected Vertices

We demonstrate the selection of vertices using both the previously proposed methods in Section 4 and the proposed fast method. The original graph is a random sensor graph with $N = 500$ nodes, and we choose $|\mathbf{S}| = M = 20$ vertices. We present sampling implementations from the same sampling distribution probability. The figure 1 illustrates the original graph signal, the reconstructed signal based on the proposed method, and the graph spectrum in the GLCT domain, where GLCT parameters are set as $\xi = 1, \beta = 2, \alpha = 0.9$. And the original graph signal is a realistic model using noise band-limited signal [17], that is

$$\hat{\mathbf{y}} = \mathbf{O}^M \mathbf{y} = \left[\hat{\mathbf{y}}_{|\mathcal{F}|}^\top \mathbf{0}_{N-|\mathcal{F}|}^\top \right]^\top + \epsilon,$$

where $\hat{\mathbf{y}}_{|\mathcal{F}|} \in \text{BL}_{|\mathcal{F}|}(\mathbf{O}^M)$ is a random vector of length $|\mathcal{F}|$, with its elements following $\mathcal{N}(0, 0.2)$, and $\epsilon \in \mathbb{R}^N$ is an independent identically distributed noise vector obeying $\mathcal{N}(0, 5 \times 10^{-3})$. For simplicity, we set $|\mathcal{F}| = |\mathbf{S}| = 20$.

The underlying graph and the sampling results are illustrated in Figure 2. Observably, *MinPinv*, *MaxSigMin*, *MaxVol*, and the proposed method select uniformly distributed sampling positions (with the proposed method being the fastest among the six). Due to the fact that *MinFro* and *MaxSig* do not consider the positions of the

already selected vertices in each iteration, they frequently choose vertices very close to the ones already selected, as illustrated in Fig. 2 (a) and (c). Consequently, this leads to significant reconstruction errors.

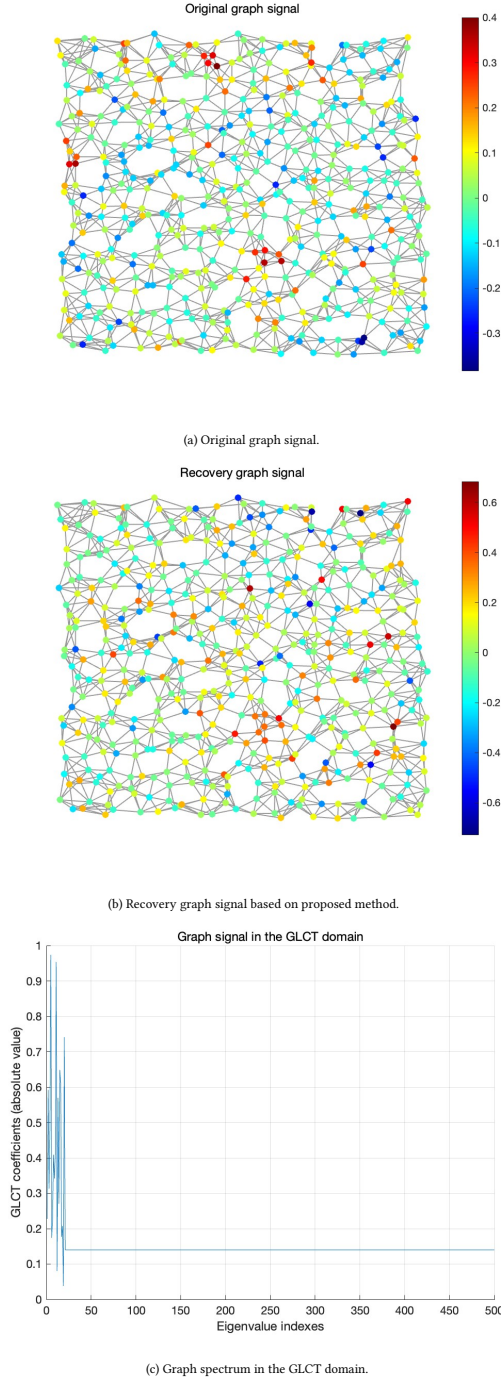


Figure 1: Graph signal representation.

Table 5: The Speed-up of Proposed Method Relative to The Previous Method

Methods	speed-up
MinFro	12.7796
MinPinv	1.2825×10^3
MaxSig	1.1937×10^3
MaxSigMin	1.1044×10^3
MaxVol	1.8097×10^3

5.2 Comparison of Recovery Time

In this section, we compare the execution time for selecting $|\mathcal{S}|$ for different $|\mathcal{V}| = N$, choosing $|\mathcal{S}| = M = N/10$ sampling vertices. Figure 3(a) depicts the comparison of execution times as N varies for the randomly generated graph with community structure. The results are presented as the average of 10 independent runs [17].

In deterministic methods, MinFro and the proposed method exhibit faster computational efficiency compared to other methods. For instance, when $N = 2642, M = 200$ for the Minnesota traffic graph, the speed-up of our method relative to the previous method is defined as

$$(\text{Time of Previous Method}) / (\text{Time of Proposed Method}),$$

and the results are summarized in Table 5. The proposed method is at least 100 times faster than methods with smaller recovery errors. While MinFro is indeed very efficient, boasting only a roughly 10-fold difference in computation time compared to the proposed method, it shows a noticeably higher recovery error than other methods, as we will delve into in the following section.

5.3 Comparison of Recovery Errors

We compared the recovery errors between the proposed fast method and previous methods, utilizing the 256-node Swiss roll graph [20]. The construction of the original graph signals follows the GLCT parameters outlined in Section 5.1. The average mean squared error (MSE) between the recovered and original signals is summarized in Figure 3(b).

Clearly, with fewer sampling nodes, MaxVol, MinPinv, and MaxSigMin exhibit the lowest MSE. However, as the number of sampling nodes increases, the proposed method achieves the lowest MSE, and the error remains stable, while other methods show unstable errors. Moreover, the reconstruction times for MinPinv, MaxSig, MaxSigMin, and MaxVol are over 1000 times longer than that of the proposed method. Although the recovery time of MinFro is closest to that of the proposed method, both the recovery time and MSE are higher than those of the proposed method. Therefore, overall, the proposed method yields the optimal results.

6 CONCLUSION

Due to the high computational complexity and longer execution time associated with previous sampling algorithms, we propose a novel sampling set selection method based on the localization filter operators in the GLCT domain. The proposed method shares close ties with the previous sampling method based on graph linear canonical frequencies, and we summarize the relationships between

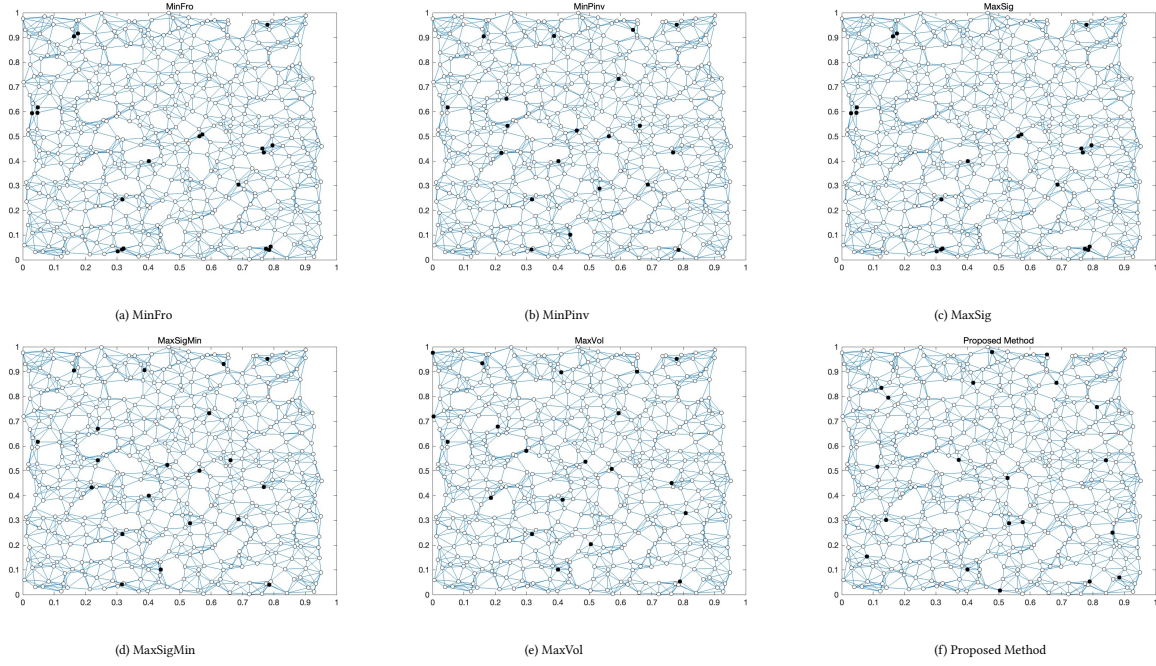


Figure 2: Sampling vertices of a random sensor graph ($N = 500$). Twenty vertices are selected, whose are marked in black.

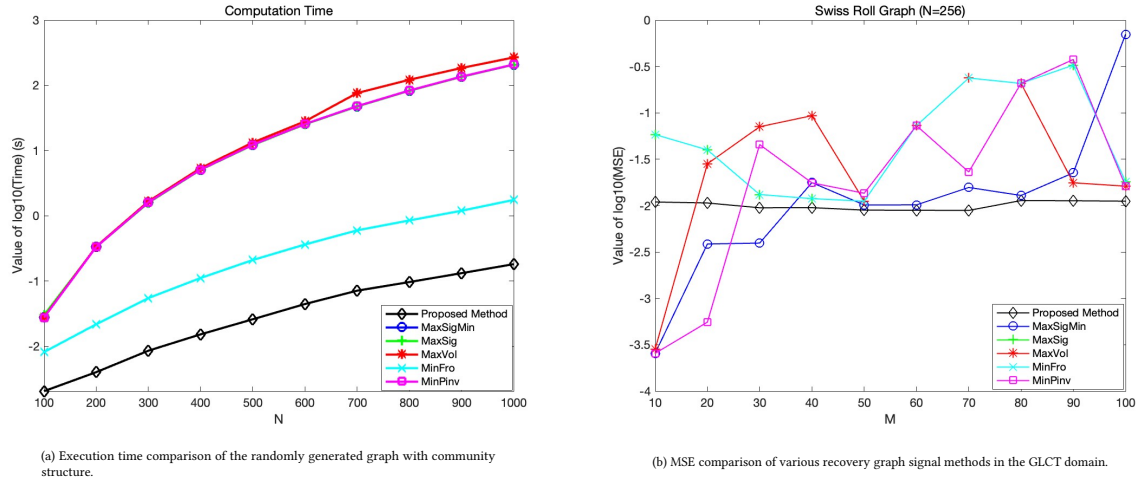


Figure 3: Comparison of recovery time and errors.

them. Notably, our proposed sampling set selection method eliminates the need for eigenvalue decomposition of the adjacency matrix in the GLCT domain, while still incorporating both frequency and vertex information from the graph. This method significantly enhances the speed of the sampling algorithm, and its performance is demonstrated to be optimal through comprehensive numerical experiments.

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REFERENCES

- [1] D. I. Shuman, S. K. Narang, P. Frossard, A. Ortega, and P. Vandergheynst, "The emerging field of signal processing on graphs: Extending high-dimensional data analysis to networks and other irregular domains," *IEEE Signal Process. Mag.*, vol. 30, no. 3, pp. 83–98, 2013.
- [2] A. Sandryhaila and J. M. F. Moura, "Big data processing with signal processing on graphs: Representation and processing of massive data sets with irregular structure," *IEEE Signal Process. Mag.*, vol. 31, no. 5, pp. 80–90, 2014.
- [3] D. K. Hammond, P. Vandergheynst, and R. Gribonval, "Wavelets on graphs via spectral graph theory," *Appl. Comput. Harmon. Anal.*, vol. 30, no. 2, pp. 129–150, 2011.
- [4] A. Sandryhaila and J. M. F. Moura, "Discrete signal processing on graphs," *IEEE Trans. Signal Process.*, vol. 61, no. 7, pp. 1644–1656, 2013.
- [5] A. Ortega, P. Frossard, J. Kovačević, J. M. F. Moura, and P. Vandergheynst, "Graph signal processing: Overview, challenges, and applications," *Proc. IEEE*, vol. 106, no. 5, pp. 808–828, 2018.
- [6] G. Leus, A. G. Marques, J. M. F. Moura, A. Ortega and D. I. Shuman, "Graph signal processing: History, development, impact, and outlook," *IEEE Signal Process. Mag.*, vol. 40, no. 4, pp. 49–60, 2023.
- [7] C. Maier and D. Simovici, "Saturated betweenness centrality sets of vertices in graphs," *J. adv. info. technol.*, vol. 12, no. 4, pp. 287–295, 2021.
- [8] D. I. Shuman, B. Ricaud, and P. Vandergheynst, "Vertex-frequency analysis on graphs," *Appl. Comput. Harmon. Anal.*, vol. 40, no. 2, pp. 260–291, 2016.
- [9] Y. Q. Wang, B. Z. Li, and Q. Y. Cheng, "The fractional Fourier transform on graphs," in *Proc. Asia-Pacific Signal Inf. Process. Assoc. Annu. Summit Conf. (APSIPA ASC)*, 2017, pp. 105–110.
- [10] Y. Zhang and B. Z. Li, "The fractional Fourier transform on graphs: Modulation and convolution," in *Proc. 8th IEEE Int. Conf. Signal Image Process. (ICSIP)*, 2023, pp. 737–741.
- [11] Y. Zhang and B. Z. Li, "Discrete linear canonical transform on graphs," *Digit. Signal Process.*, vol. 135, 103934, 2023.
- [12] H. M. Ozaktas, Z. Zalevsky, and M. A. Kutay, "The fractional Fourier transform with applications in optics and signal processing," *Wiley Sons*, Chichester, New York, 2001.
- [13] Y. Zhang and B. Z. Li, "Discrete linear canonical transform on graphs: Uncertainty principles and sampling," *Submitted for publication. Our submission was on Jan. 24, 2024, 2024*.
- [14] S. Chen, R. Varma, A. Sandryhaila, and J. Kovačević, "Discrete signal processing on graphs: Sampling theory," *IEEE Trans. Signal Process.*, vol. 63, no. 24, pp. 6510–6523, 2015.
- [15] M. Tsitsvero, S. Barbarossa and P. D. Lorenzo, "Signals on graphs: Uncertainty principle and sampling," *IEEE Trans. Signal Process.*, vol. 64, no. 18, pp. 4845–4860, 2016.
- [16] A. Anis, A. Gadde and A. Ortega, "Efficient sampling set selection for bandlimited graph signals using graph spectral proxies," *IEEE Trans. Signal Process.*, vol. 64, no. 14, pp. 3775–3789, 2016.
- [17] A. Sakiyama, Y. Tanaka, T. Tanaka, and A. Ortega, "Eigendecomposition-free sampling set selection for graph signals," *IEEE Trans. Signal Process.*, vol. 67, no. 10, pp. 2679–2692, 2019.
- [18] S. C. Pei, Y. C. Lai, "Discrete linear canonical transforms based on dilated Hermite functions," *J. Opt. Soc. Am. A*, vol. 28, no. 8, pp. 1695–1708, 2011.
- [19] N. Perraudin, B. Ricaud, D. I. Shuman, and P. Vandergheynst, "Global and local uncertainty principles for signals on graphs," *APSIPA Trans. Signal Inf. Process.*, vol. 7, p. e3, 2018.
- [20] N. Perraudin, J. Paratte, D. Shuman, V. Kalofolias, P. Vandergheynst, and D. K. Hammond, "GSPBOX: a toolbox for signal processing on graphs," 2016, arXiv preprint, arXiv: 1408.5781.